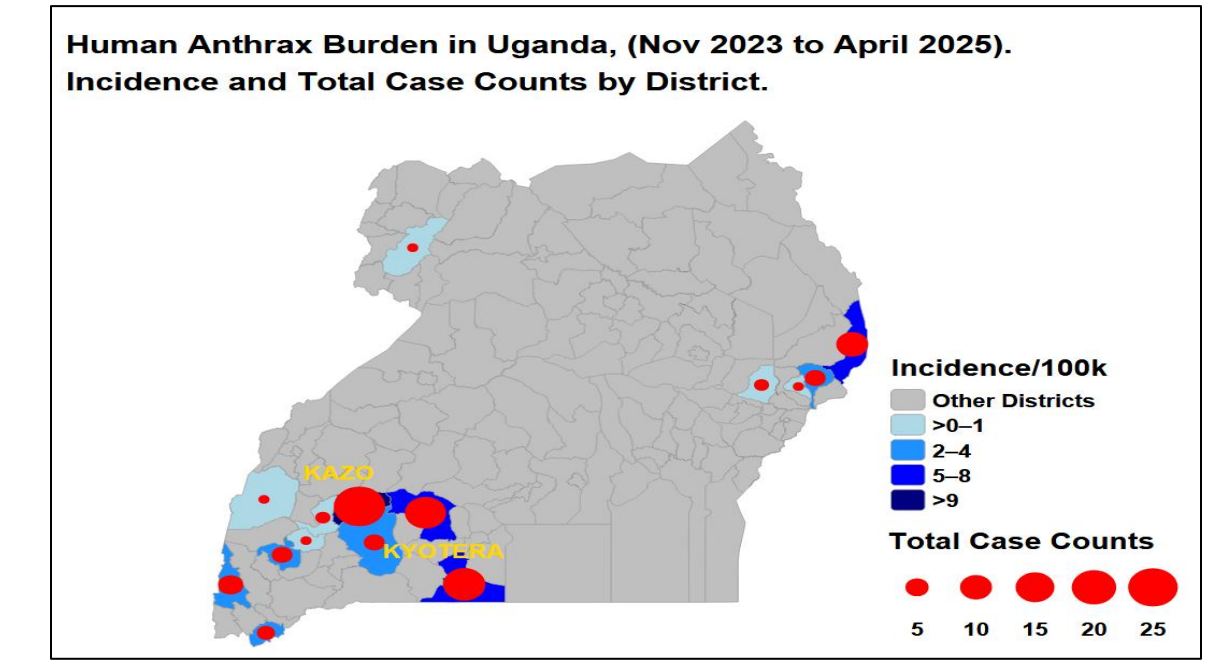
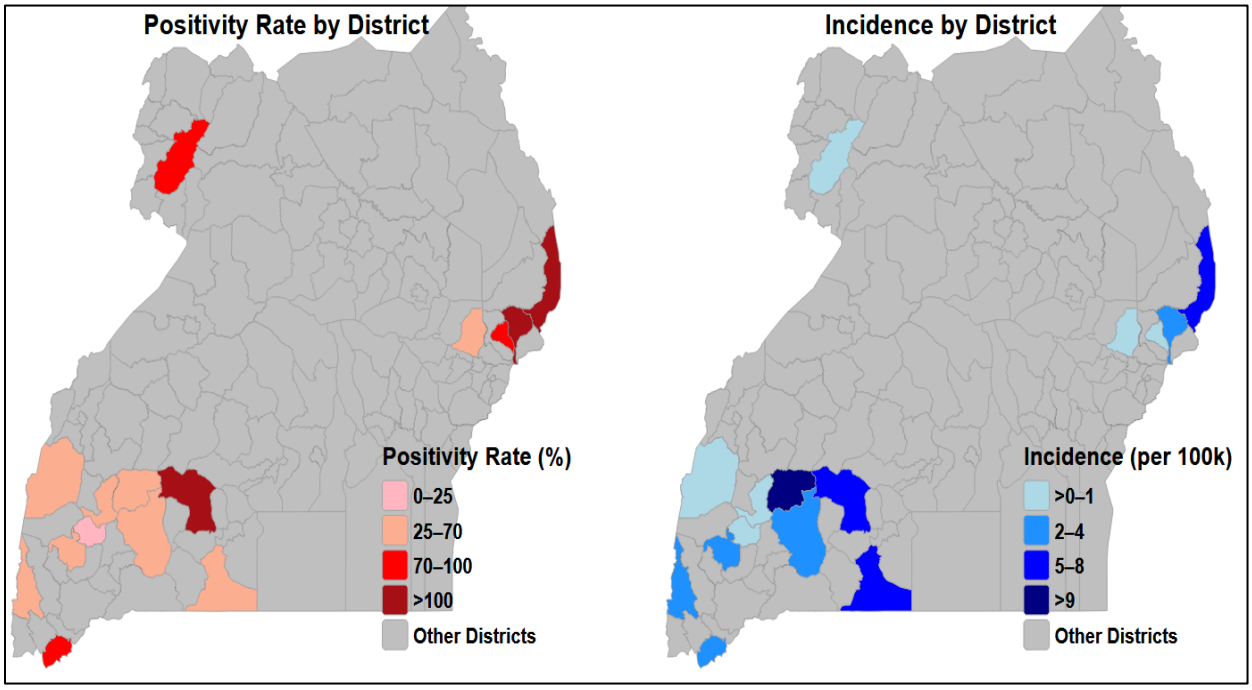
Spatial clustering of Anthrax in Uganda (Nov 2023 to April 2025) and implications for targeted surveillance.



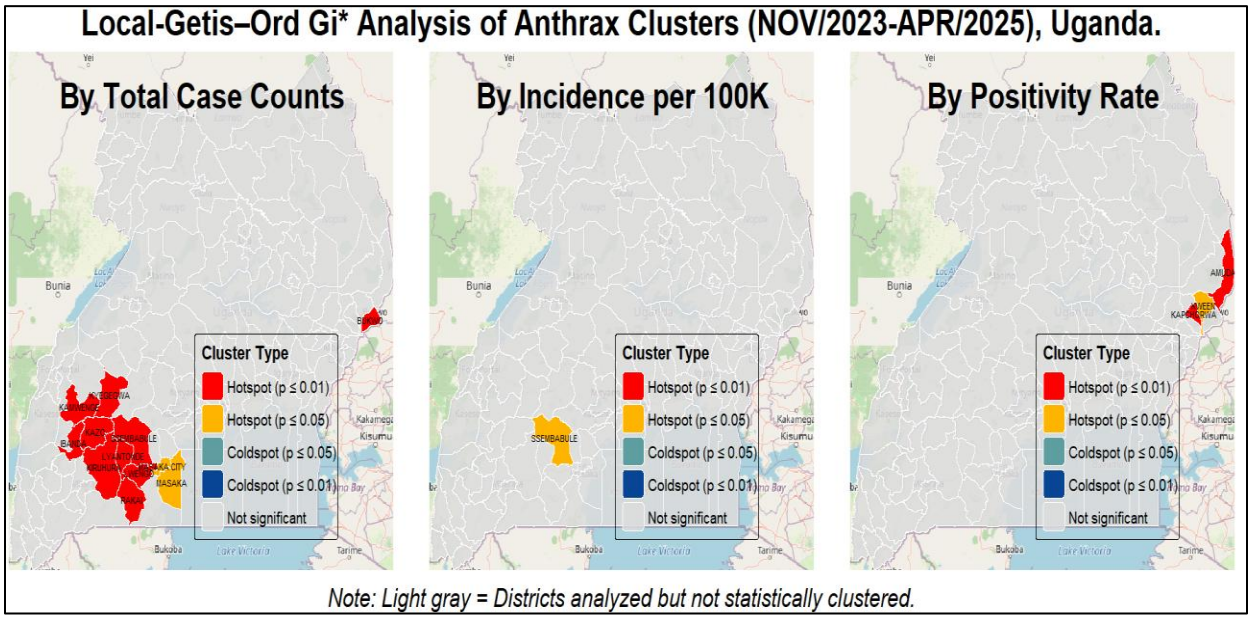


Gray represent districts that reported no anthrax cases.

Districts labeled in yellow; **Kazo** and **Kyotera** registered the highest case counts of Anthrax during the observed study period. Border Districts in Eastern Uganda, show high incidence rates, as seen from the deep blue shading combined with large red circles.

The maps reveal higher incidence concentrations in Southwestern Uganda, while elevated positivity rates appear in both the Southwest and Eastern regions. In some Eastern districts positivity rates are high but reported incidence is low, such could suggest weaker surveillance system i.e., suboptimal active case search and limited laboratory diagnostic testing mostly subjected to those with strong sign and symptoms.

To gain a deeper understanding of the geographic distribution, spatial analysis results were derived from applying both the Global Moran's I and Local Getis-Ord Gi* statistical tests to identify and distinguish geographic districts with significant spatial clustering of the reported cases and patterns are presented below.



Districts with statistically significant Anthrax clusters are shaded in Yellow and Red colors.

Total case counts

The spatial clustering of anthrax case counts is statistically significant (Hotspot at $p \leq 0.01$) in southwestern districts of Kyegegwa, Kiruhura, Kazo, Lwengo, Lyantonde, Sembabule, Ibanda, Kamwenge, and Rakai as well as Central Uganda districts (Hotspot at $p \leq 0.05$) Masaka and Masaka City. These results indicate a high burden zone in Western/Central Uganda. These districts likely share common exposure patterns that warrant further evaluations.

There are no coldspots detected as no districts met coldspot thresholds. This absence could imply wide stake similaries with either similar low burden or random spatial distribution in non-hotspot zones.

Interestingly, some districts like **Bukwo**, **Rakai**, and **Lyantonde**, although reporting zero or no Anthrax cases during the observed period (Nov 2023 to April 2025), were statistically classified as part of significant hotspot clusters at 99% CI for case counts. This can arise due to their close proximity to high case neighboring districts. This is because the GI* statistic doesn't evaluate a location in isolation but how values in a local neighborhood differ from the global mean as well

as spillover effect from the spatial weights. A district with zero cases might still be surrounded by districts with very high values, and this spatial association can yield a significant hotspot result due to the influence of neighbors.

Incidence rate hotspot (per 100K population)

Only **Ssembabule** as a hotspot. This suggests that Ssembabule appears as a priority area despite not having the highest case count.

Positivity rate hotspot

From the right map panel; **Amudat, Kapchorwa** and **Kween** are shown as significant positivity rate hotspots. The eastern districts, especially along the Karamoja Kenya border, have high test positivity, which could suggest under testing with selective testing of likely cases, or delayed response in surveillance activities. Such districts may represent surveillance blind spots.

Remarks

These findings underscore the importance of not solely relying on raw case numbers but integrating spatial influence from neighboring areas in outbreak detection.

These results support targeted surveillance, resource allocation, in districts exhibiting clustering, regardless of whether local counts are high or not.